

Cover Page

Declaration

Abstract

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1. Introduction

The consumer journey for vehicle purchasing has changed significantly. Nowadays, a search query or a quick glance through a social media feed sparks what used to be a physical showroom visit or a print magazine review. Social media platforms have turned into multi-purpose location, effectively functioning as a community hub, a 24/7 customer support desk and modern showroom. This shift demands that automotive brands adopt an elaborate, subjective technique to marketing in place of just promoting their products.

Connecting with younger consumers whose purchasing intentions are based on passion and community rather than just business is significant challenge for the automotive industry [1]. Recent trends show a noticeable change, with younger generations being more interested in owning classic, sports and collector cars than their predecessors [2]. The reason they actively participate in unique subcultures like classic car restoration is to spark interest, passion, and curiosity among the younger generation, including ladies, through sharing videos and pictures of such restored vehicles on social media [3] [4]. Unfortunately, social representations regarding classic car modification and restoration create unique marketing opportunities that remain highly untapped in Sri Lanka [5].

A significant research gap exists in how these automotive brands currently use social media platforms to interact with car enthusiasts [6] [7]. While the automotive industry is growing globally with the adaptation of digital marketing, limited research exists on the Sri Lankan context, specially focusing on young car enthusiasts and restoration parties which perform poorly on platforms like Instagram, Facebook, TikTok and YouTube, are still failing to unlock its full potential through social representations [8] [9].

The research aims to address the key necessity to understand how social media can effectively engage young automotive enthusiasts in Sri Lanka's classic and sports car modification and restoration subculture, while increasing the customer base, to be a spark in classic, sports car modification and restoration in Sri Lanka, to boost enthusiasm and build a talented team as a brand internationally in Sri Lankan automotive industry for the restoration of vintage sports cars.

The study will offer useful frameworks for local workshops, by unlocking the digital transformation in emerging market automotive industry.

1.1 Background and Literature Survey

The early research focused on the shift from push marketing to pull marketing using static websites and search engine optimization. The subsequent background, which forms the State-of-the-Art today, involves the analysis of user-generated content (UGC) and social media engagement (likes, shares, comments) on platforms like TikTok, Facebook, WhatsApp, YouTube and Instagram. State-of-the-art techniques primarily involve sentiment analysis of social media comments to gauge brand perception and econometric modeling to correlate social media campaign spending with sales figures.

The literature highlights two critical trends shaping the automotive sector:

1. The Digital Consumer Journey

The traditional, linear consumer journey is now fragmented and digitally initiated. A search query or a social media post often replaces the initial physical showroom visit. Social media's influence on purchase decisions is well-documented, with platforms like Instagram and YouTube driving discovery and product evaluation, especially for emotionally charged purchases like customized vehicles. The literature suggests that brands must adopt an elaborate, subjective technique to marketing that prioritizes storytelling, authenticity, and community interaction over simple product promotion.

2. The Rise of Subculture-Driven Consumption

Recent trends confirm a noticeable change in youth interests, showing an increased participation in, and ownership of, classic, sports, and collector cars. This engagement is characterized by active participation in unique subcultures (e.g., JDM, muscle car restoration). For businesses operating in this niche (e.g., modification workshops, restoration service providers), success is dependent on being an *active, credible member* of the community, not just a seller. This requires sophisticated content strategies and digital tools capable of hosting and nurturing such niche interests.

1.2 Research Gap

While the literature confirms the shift to digital consumption and the growing importance of subcultures, a significant gap remains in the localized and specialized application of this knowledge. Specifically:

1. **Lack of an IS Framework for Niche Local Businesses:** There is a lack of structured, information system-based frameworks tailored to help small-to-medium-sized local automotive modification and restoration service providers effectively leverage social media engagement to drive business outcomes (e.g., lead generation, trust building, and purchase intent) within a specific geographic context.
2. **Quantifying Subjective Engagement:** The existing metrics often fail to capture the subjective element of purchase intent—the passion and community loyalty—which is the primary driver for this target demographic.

This project addresses this by proposing an Information System Engineering solution designed specifically to model, measure, and enhance the digital interaction within the classic/sports car enthusiast subculture.

Comparative analysis with past researches

Research Dimension	Past Research	This Study	Novel Contribution
Geography	USA, Europe, Australia	Sri Lanka	First developing market study
Industry	New vehicle sales	Restoration/modification	Niche market focus
Demographics	General consumers	Ages 18-35 youth	Youth-centric strategies
Platforms	Facebook-centric	Instagram, YouTube, TikTok, Facebook	Multi-platform comparison
Methods	Single-method	Mixed-methods + experimental	Methodological rigor
Business Size	Large corporations	SMEs (<5 employees)	Resource-appropriate solutions
Technology	Enterprise solutions	SME-appropriate IS design	Accessible technology
Validation	Theoretical only	Real business A/B testing	Proven effectiveness
Culture	Western context	Sri Lankan cultural grounding	Cultural adaptation
Knowledge	Academic papers	Implementation guides + training	Practitioner accessibility

1.3 Research Problem

The research problem is:

How can a local automotive restoration and modification business effectively leverage social media and community-focused digital platforms to convert the passion-driven engagement of young car enthusiasts into measurable, localized customer acquisition and service patronage?

2. Objectives

2.1 Main Objective

To develop an effective social media engagement framework that increases young automotive enthusiasts in Sri Lanka's classic and sports car modification and restoration market.

2.2 Specific Objectives

- To quantify and analyze foundational concepts of social media engagement in automotive restoration communities.
- To profile the young automotive enthusiasts (ages 18-35) in Sri Lanka and their digital subcultures.
- To analyze the role of major social media platforms in engaging automotive subcultures.

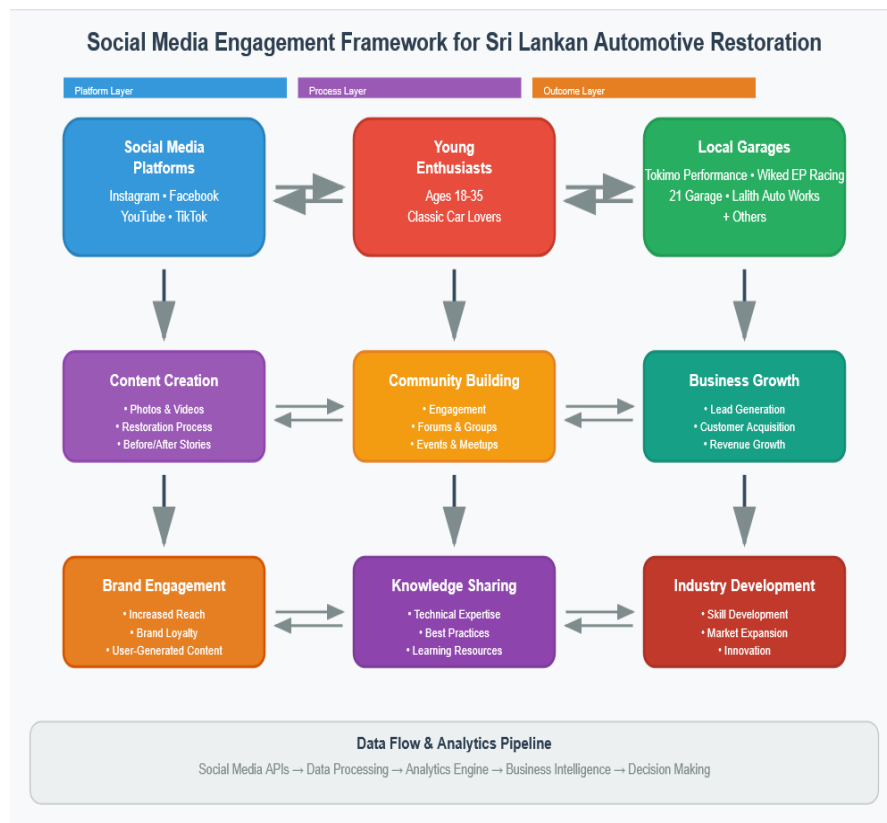
3. Methodology

3.1 Overall System Diagram

The proposed solution involves to develop an effective social media engagement framework that increases young automotive enthusiasts in Sri Lanka's classic and sports car modification and restoration market. This approach will bridge the gap between traditional automotive marketing and community-centric and digitalized emerging marketing engagement.

Solution Includes:

1. Digital culture study: Analysis of existing social media communities and subcultures.
2. Platform- specific engagement techniques: Personalized methods for Instagram, Facebook, YouTube, and TikTok
3. Community-centric marketing approach: Moving from advertising to actively participating in marketing.
4. Sri Lankan garage integration framework: Connecting local restoration and modification workshops by linking online communities.



The solution will create a system where social media posts about classic car modification and restoration attract and engage car enthusiasts in Sri Lanka. This community engagement will help local businesses grow and allow young generation to learn and share their expertise.

By understanding the interest and how young car enthusiasts use social media, the framework will enable more effective in the car modification and restoration subculture in Sri Lankan automotive industry.

3.2 Tasks and Subtasks

1) Literature review completion

- Analysis of existing automotive social media strategies.
- Identification of engagement patterns in restoration parties.
- Finalize background data analysis.

2) Survey design and collect data

- Design and data collection form for young enthusiasts.
- Finalize questionnaires
- Deploy survey to target group (Age 18-35)

3) Data analysis and modeling

- Clean Data, conduct statistical analysis.
- Platform-specific content analysis.
- Design database

4) Tech stack setup

- Initialize front-end and back-end services.
- Design the system
- Core feature development

5) User Acceptance Testing (UAT) and Documentation

- Test the system
- Create UAT report
- Finalize the report

3.3 Tools and Techniques

- Equipment and support : High performance PC/Laptop , Internet connection, Code editor (VS code)
- Software / Tools : Front-end (React, CSS) , Back-end (Firebase), Analysis (R Studio)
- Data Collection Tools : Google Forms

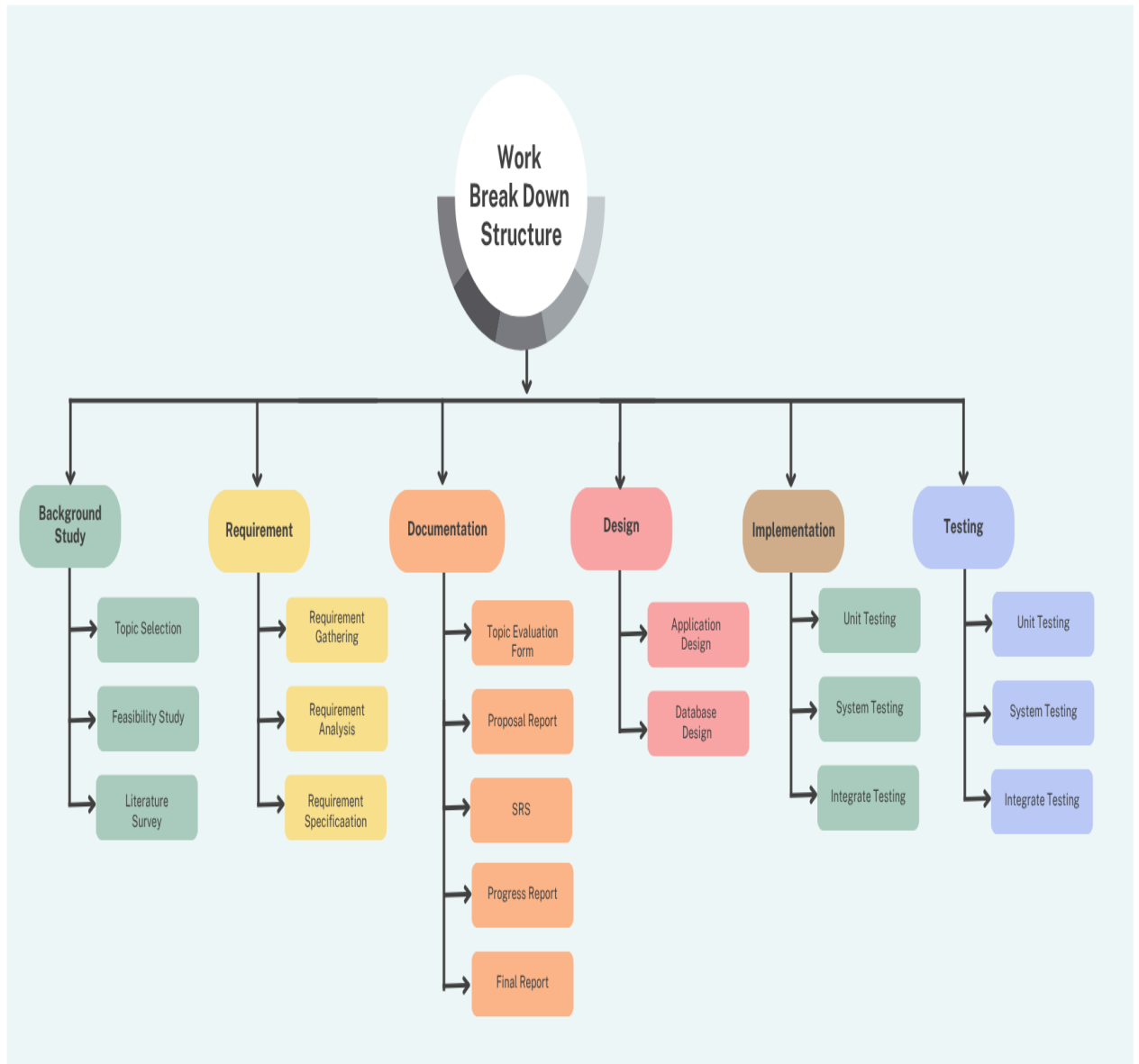
3.4 Data Collection

Data will be collected via survey responses, interviews, physical visits, social media posts, social media groups.

Data will be collected from local garage owners, young car enthusiasts, vehicle buyers, car sale owners, other automotive enthusiasts, social media users, content creators.

1. **Digital Marketing:** Need experts who understand how to use social media to build communities. This includes knowing social media algorithms, what content works for young car enthusiasts, and how to use data to track younger generation interest in online. Need to know how to collaborate with local influencers.
2. **Automotive Industry:** Need a deep understanding of car restoration is critical, including the specific terms and processes. Need to familiar with the Sri Lankan automotive market, how local garages operate, and the subcultures of car enthusiasts.
3. **Research:** Will use a mixed-methods approach, combining surveys to gather data from many people with interviews and focus groups to get detailed insights from garage owners and car enthusiasts.
4. Project will use both **primary and secondary data**. Primary data will be collected directly from the audience through surveys and interviews. Secondary data will be collected from existing social media analytics and industry reports.
5. Will use a variety of **technical tools** to process the information, including social media analytics software, survey platforms, and programs for statistical analysis.
6. Will build a strong **collaborative network** by partnering with local garages and young car enthusiast groups. This ensures the research is relevant and the strategies are practical for the real world.

3.5 Work Break Down Structure



4. Project Requirements

4.1 Functional Requirements

4.2 Non-Functional Requirements

4.3 User Requirements

4.4 System Requirements

4.5 Test Cases

5. Budget

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7. Appendix